

**COMPUTER SCIENCE MAJOR
CLASS OF 2027**



ACADEMIC DOSSIER

**OFFICE OF UNDERGRADUATE EDUCATION &
ACCREDITATION**

Table of Contents

PROGRAM DESCRIPTION:	3
PROGRAM EDUCATIONAL OBJECTIVES:	3
PROGRAM LEARNING OUTCOMES:	4
MAPPING OF PLOS TO UNIVERSITY LEARNING GOALS:	5
REQUIRED COURSES:	6
LIST OF ELECTIVES:	12
GRADUATING REQUIREMENTS:	12
OPTION FOR PURSUING MINOR(S):	14
4-YEAR GRID:	15

#	Revision Description	Date Revised	Approved By	Date Approved
1.4	Updated list of electives	05/02/2025	N/A	N/A
1.3	Revision in 4-year grid and graduating requirements	04/07/2024	AC	18/09/2024
1.2	Revised 4-year grid and graduating requirements	21/03/2024	BoF	03/04/2024
1.1	Updated list of electives	13/02/2024	N/A	N/A
1.0	New document	N/A	AC#22	12/09/2023



COMPUTER SCIENCE

PROGRAM DESCRIPTION:

Computer Science (CS) is the study of computation - what can and cannot be computed, how computation can be made more efficient, how to build machines that can compute, and which spheres of human activity can benefit from computational approaches. It is deeply rooted in logic and mathematics. Theoretical Computer Scientists push the frontiers of computation by inventing new computational approaches. Practical Computer Scientists apply the theory of Computer Science to different application areas like science, finance, medicine, business, transportation, entertainment, education, communication, engineering, art, and the humanities.

Interventions stemming from CS are just beginning to disrupt and reinvent Pakistani society. The CS program provides students the intellectual and technical foundation to assess these interventions and to contribute meaningfully and thoughtfully to the transition of our society to the information age. With an education grounded in the Liberal Arts, our graduates have an unrivalled understanding of our society and the ethical ramifications of technology.

A major with the program educates students in the theory, systems, and applications of CS so that they are able and willing to make impactful contributions to society and are prepared for success in industry, entrepreneurship, and higher education.

PROGRAM EDUCATIONAL OBJECTIVES:

The Computer Science program at Habib University aims to produce competent computer scientists who possess the following qualities:

1. **Technical Excellence:** Our graduates will have a strong foundational knowledge of mathematics and computer science, along with accompanying skills both in breadth and depth. This will allow them to position themselves equally well in the Information Technology industry, as technology entrepreneurs, or in graduate programs in Computer Science or other technical and scientific fields.
2. **Life-Long Learning:** Our graduates will have a hands-on approach to self-learning and research, continually enriching their knowledge, skills, and technical know-how.
3. **Ethical Action:** Our graduates will contribute to society in an ethical manner, with an appreciation of the historical, social, and philosophical context.
4. **Effective Communication:** Our graduates will effectively communicate and collaborate with people from diverse backgrounds and in a variety of settings.

PROGRAM LEARNING OUTCOMES:

Upon graduation, students will have the following abilities:

1. Theoretical Computer Science: recall and apply foundational principles of computer science.
2. Application Development: build software systems of varying complexity in light of fundamental computer science principles and any other constraints.
3. Analysis and Design: perform technical analysis and design using core computing and mathematical knowledge.
4. Systems: apply the knowledge of computing systems.
5. Research and Exploration: develop expertise in and contribute to a given sub-field of computing by drawing upon a strong foundation in the fundamentals of computer science and mathematics to solve real life problems.
6. Problem Solving: identify and analyze problems and propose effective computing-based solutions.
7. Practical Exposure: make effective use of current tools, technologies, and good industry practices.
8. Responsible Citizenship: conduct their computing practice in a manner that is ethical and socially responsible, and corresponds to their distinct sense of identity and service to the community.
9. Self-Learning: continuously adapt their skills to the changes taking place around them.
10. Design Thinking: apply design thinking principles to the design of a solution.
11. Multi-disciplinarity: incorporate knowledge and input from multiple disciplines.
12. Communication and Teamwork: communicate and function effectively as a member or a leader of a variety of teams.

MAPPING OF PLOS TO UNIVERSITY LEARNING GOALS:

PROGRAM LEARNING OUTCOMES - MAPPING											
			University Learning Goals (ULG)								
			Know			Act			Value		
			Knowledge	Interdisciplinary & Transdisciplinary	Context	Creativity & Innovation	Critical Inquiry	Communication & Collaboration	Social Impact	Thought Self-Cultivation	Ethical & Cultural Competence
			Breadth & Depth	Synthesis & Connections	Contextually Grounded	Imaginative & Interesting	Analysis & Critical Thought	Interaction & Teamwork	Service & Sustainability	Yohsin Values & Lifelong Learning	Personal & Professional Ethics
			ULG 01	ULG 02	ULG 03	ULG 04	ULG 05	ULG 06	ULG 07	ULG 08	ULG 09
PLO	Title	Statement	Demonstrate both a genuine breadth of knowledge through the Habib Liberal Core and a capable depth of knowledge through command of their chosen major.	Synthesize knowledge, methods and viewpoints from different disciplines to both make meaningful connections among and transcend them.	Demonstrate their knowledge is grounded in a firm understanding of the historical, social, political, economic, religious, regional and global contexts in which they are located.	Imagine, develop and produce creative, original ideas, interpretations and works.	Analyse and formulate relevant critical questions, and answer those questions in a substantive way supported by quantitative and qualitative evidence.	Listen actively to comprehend the meaning of others and successfully express cogent meaning through capable oral, written, and artistic modes of communication. Effectively interact and collaborate with others.	Recognize the reciprocity of knowledge and service, and benefit their community, society and the environment through socially responsible and sustainable engagement.	Cultivate lifelong curiosity by engaging in inquiry and reflection to acquire and apply new knowledge.	Develop and nurture their own beliefs, values and sense of responsibility to reach informed conclusions, while considering, appreciating and respecting the perspectives of others.
1	Theoretical Computer Science	recall and apply foundational principles of computer science.	S		W		W				
2	Application Development	build software systems of varying complexity in light of fundamental computer science principles and any other constraints.	S	W		S		W	W		
3	Analysis and Design	perform technical analysis and design using core computing and mathematical knowledge.	S			W	W				
4	Systems	apply the knowledge of computing systems.	S	W			W				
5	Research and Exploration	develop expertise in and contribute to a given sub-field of computing by drawing upon a strong foundation in the fundamentals of computer science and mathematics to solve real life problems.	S	W		M	S	S	W	W	
6	Problem Solving	identify and analyze problems and propose effective computing-based solutions.	M	M		S	W	W	W		
7	Practical Exposure	make effective use of current tools, technologies, and good industry practices.	M		W				W	W	
8	Responsible Citizenship	conduct their computing practice in a manner that is ethical and socially responsible, and corresponds to their distinct sense of identity and service to the community.			M				S		S
9	Self-Learning	continuously adapt their skills to the changes taking place around them.	W		M		W		W	S	
10	Design Thinking	apply design thinking principles to the design of a solution.	W	M	W	S	W	W		W	
11	Multi-disciplinarity	incorporate knowledge and input from multiple disciplines.		S	W	S	W	S	W	M	
12	Communication and Team Work	communicate and function effectively as a member or a leader of a variety of teams.		M				S			
									Legend Competencies		
									S: Strongly - Program learning outcome strongly maps to the university learning goal.		
									M: Moderate - Program learning outcome moderately maps to the university learning goal.		
									W: Weakly - Program learning outcome weakly maps to the university learning goal.		
									Corelation		

REQUIRED COURSES:

CS 101 - Algorithmic Problem Solving

Credit Hours: 3-1

Pre-requisite: None

Fulfills: CS Foundation, and Formal Reasoning FoT requirement under the Habib Liberal Core

This course aims to equip you with the skills to convert your ideas into executable code. Some of these skills are algorithmic thinking and computer programming in a high-level programming language. Ultimately, for more interesting and involved ideas, one needs to delve into mathematics, know about data structures, and reason about algorithms, all in order to write more sophisticated code. All of these are covered later in the CS curriculum. But we will see in this course that there is a lot that we can make computers do with basic creativity alone.

MATH 101 - Calculus I

Credit Hours: 3-0

Prerequisites: None

Fulfills: Mandatory Math requirement for all DSSE students

This course fulfills a foundational mathematics course requirement for the Electrical Engineering, Computer Engineering and Computer Science majors. It is also a mandatory requirement for all non-DSSE students wishing to pursue a Mathematics Minor. The course lays the foundations for students to think visually, symbolically and numerically on the two overarching concepts of Differentiation and Integration. The course seeks to pave the way for students to develop the necessary computational and analytical skills (both in context and abstract terms) required in higher mathematics courses. The mode of instruction mostly follows a collaborative model where students are required to work in groups to discuss key Calculus ideas (that might be familiar) to deepen their understanding.

Natural Science Elective

The development of scientific method and analysis is a fundamental feature of modernity and its forms of knowledge. A natural science requirement is standard in higher educational and liberal institutions.

To ensure the scientific literacy of all our graduates, students pursuing a CS major are required to complete 02 Natural Science courses, one with lab component and the second could be a course with or without lab component.

PLAY 113 - Design Your Habib Experience

Credit Hours: 0-1

Prerequisite: None

Fulfills: Design requirement for CS Majors

This course will take you through the basics of Human-Centered Design: an approach to problem-solving that involves empathizing with people; defining problems; generating ideas; prototyping solutions; and testing to learn what works and what doesn't. The first twelve weeks of the course will take students through the design process, practically applying key methods and mindsets to tackle problems around us at Habib. For example, this could be redesigning the university food experience, or designing a new student governance model. In the last three weeks of the course, students will apply their skills and knowledge to a more personal challenge: designing their own Habib experience.

CS/MATH 113 - Discrete Mathematics

Credit Hours: 3-0

Prerequisite: None

Fulfills: CS Foundation

Preparation in the mathematical tools required for the future study of computer science, especially theoretical computer science, with an emphasis on logic and proofs; Topics include: sets and Venn

diagrams; propositional and predicate logic; direct and indirect proofs; mathematical induction; relations and functions; graphs; combinatorics.

CS 102/CE 171 - Data Structures and Algorithms

Credit Hours: 3-1

Pre-requisite: CS 101 – Algorithmic Problem Solving

Fulfills: CS Foundation

Motivates the design of algorithms by exploring various algorithms for a single task: linear search and binary search, bubble sort, insertion sort, selection sort, merge sort, quick sort; introduces techniques to reason about and compare algorithms: asymptotic analysis and notation, Master theorem; introduces frequently used data structures: list, tree, graph, stack, queue, hash tables; discusses and analyses basic operations on the data structures: infix, postfix, and prefix traversal, breadth-first and depth-first search, computation of graph and trees.

MATH 102 - Calculus II

Credit Hours: 3-0

Prerequisites: MATH 101 - Calculus I

Fulfills: Mandatory Math requirement for all DSSE students.

Calculus is a very important part of Applied Mathematics which in turn serves as an important tool in Science and Engineering. In Calculus I you studied the fundamental concepts of function and the 4 techniques of differentiation and integration. Calculus II builds upon the concepts of calculus learned in the previous course and extends them to other areas of Applied Mathematics such as multivariable functions and vectors.

CS 130/EE 172/CE 222 - Digital Logic and Design

Credit Hours: 3-1

Prerequisites: None

Fulfills: This course meets program requirements for CS, EE, and CE Majors.

Digital Electronics is at the very heart of the modern world. From smart phones to ‘Big Data’ to the electronics that controls our everyday lives, digital electronics plays a central role. An understanding of the basics of digital logic design is an essential skill for every engineer irrespective of their field of interest.

CS 355/CE 373 - Database Systems

Credit Hours: 3-1

Prerequisite: CS 102/CE 171 - Data Structures and Algorithms

Fulfills: CS Kernel

This course explores in detail the theoretical and practical aspects of Relational Database Management Systems (RDBMS) and builds an understanding of database modeling, relational algebra, structured query language (SQL), transaction management, and concurrency control. The course also covers topics involving database connectivity with front-end applications and discusses administrative aspects of database systems including database security, database management vs data warehousing vs data mining, and big data and its challenges. The modern database techniques are also part of the course and it discusses MongoDB as one of the modern databases.

CS 224/CE 272 - Object Oriented Programming and Design Methodologies

Credit Hours: 3-1

Prerequisite: CS 102/CE 171 – Data Structures and Algorithms

Fulfills: CS Kernel

Introduces object oriented and related memory concepts; motivates C++ as the language of choice; topics include: pointers and structs, objects, heap allocation, data encapsulation, classes, namespaces, constructors and destructors, virtual functions and destructors, operator overloading and standard input/output, inheritance and polymorphism, templates, standard library containers, and software design using UML 2.0.

MATH 202 - Engineering Mathematics

Credit Hours: 3-0

Prerequisites: MATH 102 - Calculus II

Fulfil: Mandatory Math requirement for all DSSE students

This course is designed to equip engineering and computer science students with some necessary mathematical tools and analytical reasoning methods that are an integral part of their future professions. Topics in the course have a wide range of applications such as, optimization problems, electrical circuit analysis, signal processing, mathematical modelling, etc.

CS 201/CE 372 - Data Structures II

Credit Hours: 3-0

Prerequisite: CS 102/CE 171 – Data Structures and Algorithms, CS 224/CE 272 – Object Oriented Programming and Design Methodologies, and CS/MATH 113 – Discrete Mathematics

Fulfills: CS Kernel

Data Structures are at the heart of all efficient software. While transparent to the user, the choice of data structure plays a crucial role in the efficiency of software. In addition, the study and design of data structures pose an interesting intellectual challenge. This course aims to deliver proficiency in the use of commonly used data structures (such as arrays, stacks, queues, Trees) and in reasoning about the trade-offs involved in the choice of a data structure. The course also introduces a few higher-level data structures (such as AVL Tree, Red-Black Tree, Heap, Tire and Inverted Index) and impart a deeper understanding of the covered data structures through implementation.

CS 330/CE 321/EE 371 - Computer Architecture

Credit Hours: 3-1

Prerequisite: CS 130/EE 172/CE 222 – Digital Logic and Design

Fulfills: CS Kernel

The course is offered under the CS Systems Requirement category. It studies the architecture of RISC-V processor that enables general purpose computing; develops hands-on expertise in developing complex logical components; topics include: instruction set architecture, addressing modes, processor design and computer arithmetic, pipelining, memory systems, fetch execution cycle, processor implementation on FPGA using Verilog HDL.

MATH 205 - Linear Algebra

Credit Hours: 3-0

Prerequisites: MATH 102 - Calculus II

Fulfills: Mandatory Math requirement for all DSSE students

This course aims to develop an appreciation of the basic concepts of linear algebra and their applications in geometry and differential equations. This will enable students to address relevant problems in the wider fields of computer science, engineering and natural sciences. The course content will be delivered mainly through in-class lectures. Students are expected to attend these lectures and understand both the course notes and the book content that is relevant to the course. A key feature of the course are weekly surprise in-class quizzes. These are meant to motivate regular engagement with the course content by the students, and will be instrumental in their performance on the same.

MATH 310 - Probability and Statistics

Credit Hours: 3-0

Prerequisites: MATH 102 - Calculus II

Fulfills: This course meets program requirements for EE, and CS Majors. And Quantitative Reasoning FoT requirement under the Habib Liberal Core

In the present world, we encounter situations where we have to make decisions on the basis of incomplete or imperfect information. The theory of probability helps provide a formal mechanism for understanding, quantifying, and dealing with uncertainty, which is ever present in our lives, pure science, or engineering applications. Simply, by uncertainty we mean the condition when outcomes or future are not completely

determined or can be captured by a deterministic function; they depend on a number of factors and perhaps just on pure chance. A lot of our present-day technologies will not be possible without an understanding of how to make decisions in presence of uncertainty. These technologies include all forms of wireless communication, servers, speech processing systems, network systems and so many more. Equipping yourself with tools to deal with uncertainty will help you with whatever you wish to pursue in life.

CS 351 – Artificial Intelligence

Credit Hours: 3-0

Prerequisites: CS 102/CE 171 – Data Structures and Algorithms, and MATH 310 – Probability & Statistics

Fulfills: CS Kernel

Artificial Intelligence (AI) is the science and engineering of making machines intelligent. This course is an introductory course of Artificial Intelligence that broadly exposes students to different areas of AI. These areas primarily include Search and Optimization, Machine Learning, Logic and Reasoning, Recommendation Systems and Natural language Processing. During the course, students work on different real-world problems and solve them by applying suitable AI techniques. The course also discusses different social and ethical issues that are emerging with recent breakthroughs in AI.

CS 212 - Nature of Computation

Credit Hours: 3-0

Prerequisite: CS/MATH 113 – Discrete Mathematics

Fulfills: CS Kernel

Develops the foundations for theoretical computer science; provides opportunities to develop rigorous mathematical arguments; engages with classical ideas from theoretical computer science; topics include: decidability, complexity, halting problem, reduction, P vs. NP, NP-completeness, deterministic and non-deterministic finite automata, Turing machines, languages, regular expressions, grammars, Cook-Levin theorem.

CS 232/CE 324 - Operating Systems

Credit Hours: 3-1

Prerequisite: CS 102/CE 171 - Data Structures and Algorithms, and CS 330/CE 321/EE 371 - Computer Architecture (CS Systems Requirement)

Fulfills: CS Kernel

The student will be taught principles of modern operating systems. In particular, the course will cover details of concurrent processes, multi-threads, CPU scheduling, memory management, file system, storage subsystem, and input/output management. This will be accomplished by integrating theory and practice through coordinated lecture and lab hours.

CS 412/CE 471 - Algorithms: Design and Analysis

Credit Hours: 3-0

Prerequisite: CS 201/CE 372 – Data Structures II, and MATH 310 – Probability and Statistics

Fulfills: CS Kernel

Develops the tools and techniques to design and analyze algorithms for correctness and efficient. Presents and analyzes individual algorithms as instances of classes of algorithms. The chosen algorithms are useful, instructive, and/or beautiful. Topics include: asymptotic notation; run-time bounds; linear recurrence relations and their solutions, including master method; a lower bound on sorting; divide and conquer; directed acyclic graphs (DAGs) and their applications, e.g. topological sorting, identifying connected components; max-flow problem and its solutions, e.g. Ford-Fulkerson method, its relation to the min-cut problem; dynamic programming; matrix-chain multiplication; longest common subsequence and related problems; shortest path in DAGs; string edit distance and related problems; knapsack problem and its variants.

CS 353/CE 374 - Software Engineering

Credit Hours: 3-0

Prerequisite: CS 224/CE 272 – Object Oriented Programming and Design Methodologies, and CS 355/CE 373 – Database Systems

Fulfills: CS Kernel, and Creative Practice FoT requirement under the Habib Liberal Core

The course introduces students with various software process models and various activities that are performed in different phases of software development. Students will carry out some of these activities in a group project to get a feel of how software development team functions to execute a development.

CS 491 - Kaavish I

Prerequisite: CS 201/CE 372 - Data Structures II, and CS 353/CE 373 – Software Engineering

Fulfills: CS Capstone

Self-directed final year project carried out under the supervision of a faculty member; emphasizes solving a real-world problem; integrates knowledge and skills accumulated over the entirety of the degree; first of a 2-part sequence.

CS 492 - Kaavish II

Prerequisite: CS 491 – Kaavish I

Fulfills: CS Capstone

Kaavish condenses students' 4 years with the CS program at Habib University into a single course in the form of a final year project. The aim of the course is to enable students to research, propose, solve, and present a practical and meaningful problem using the learned skills and concepts in an allocated time frame.

Professional Practices Category Courses:

The students are required to enroll in either one of the following courses falling under the Professional Practices category:

MGMT 304 - Fundamentals of Intellectual Property

Credit hours: 3-0

Course Prerequisites: None

This course will cover a range of basic concepts related to intellectual property and its management, and highlight some of these by discussions of real-world case (law) examples. Other than the subgeneric (of its own kind) systems, the course content can broadly be divided into four general categories of IP: i) Patents, ii) Copyrights, iii) Trademarks, and iv) Trade Secrets.

MGMT 320 - Principles of Management

Credit Hours: 3-0

Course Prerequisites: None

The course on Principles of Management for Tech Professionals introduces management as a discipline and process to tech professionals. This course includes evolution and scope of management, decision-making, planning, strategy, organizing, staffing, leading, control, change, and the importance of management in the global environment and ethical considerations of management decisions. Hence, the course provides a framework that will enhance a person's effectiveness in the business.

MGMT 321 - Engineering Project Management

Credit Hours: 3-0

Course Prerequisites: None

Through using textbook, discussions, assignments and real-world examples, the engineering professionals will learn how to identify, define, plan, execute, monitor, control, and close projects. They will build project components, organize work efficiently, effectively and help them to control changes. The students will use tools to build works schedules, allocate resources and manage cost of any project. This will help them to get a hands-on training of using project management tools for the smooth flow of various stages of project that is the need of time and most demanded skill by the employers all around the world.



MGMT 322 - Operations Management

Credit Hours: 3-0

Course Prerequisites: None

It has become increasingly important for tech professionals to learn, manage and control work, time and resources more efficiently and effectively. This course provides the basics to teach the tech professionals about the key concepts and tools employed to design, analyze, control and improve business processes by the principles of Operations Management, the production of goods and services. Briefly, this course will prepare students to understand and apply the concepts such as bottleneck measurement, process improvement and synchronization, process measurement, service improvement, six sigma approaches, lean management, statistical quality control techniques and decision trees. Moreover, with an increase in global competition, the survival of a firm in the market more largely depends on how well it integrates the operations into its business model. Thus, it is essential for all students and potential business managers to acquire an understanding and appreciation of operations management. This will help them to acquire most demanded skill by the employers all around the world. Therefore, this course will essentially help to inculcate skills to produce well rounded tech professionals.

MGMT 323 - Supply Chain Management

Credit Hours: 3-0

Course Prerequisites: None

The course aims to provide an understanding of fundamental concepts of supply chain management. All functional areas of supply chain management are explored in an integrated view of procurement, manufacturing and operations management, transportation and logistics, inventory and warehousing, demand planning, scheduling, network design, collaboration, and performance measurement. Topics may also cover supply chain financial metrics, strategy, and risk management for demand-driven value networks.

LIST OF ELECTIVES:

Below is the proposed list of elective courses offered by the CS Major. These elective courses may be subject to change.

Fall 2023

CS 313 - Computational Complexity Theory
 CS 316L - Introduction to Deep Learning Lab
 CS 316 - Introduction to Deep Learning
 CS 440 - Computer Graphics
 CS 446 - Applied Digital Image Processing
 CS 314/PHY 300 - Quantum Computing
 CS 370 - Web and Mobile Development
 CS 351 - Artificial Intelligence
 CS 361/SDP 35 - Social Network Analysis
 CS 457 – Data Science

Spring 2024

CS 451 - Computational Intelligence
 CS 432 - GPU Accelerated Computing
 CS 371 - Software Security
 CS 333/DES 327 - Introduction to Human Computer Interaction
 CS 335 - Introduction to Large Language Models
 CS 334 - Competitive Programming
 CS 343 - Graph Data Science

Fall 2024

CS 452 – Probabilistic Graphical Models
 CS 361/SDP 352 – Social Network Analysis
 CS 313 - Complexity Theory
 CS 316/CE 365 – Introduction to Deep Learning
 CS 316L/CE 365L – Introduction to Deep Learning Lab
 CS 457/CE 464 – Data Science
 CS 370 – Web and Mobile Development
 CS 314/PHY 300 – Quantum Computing
 CS 458/CE 463 – Natural Language Processing

Spring 2025

CS 435 – Generative AI: Security, Ethics and Governance
 CS 318 – Enterprise Software Development
 CS 370 – Web and Mobile Development
 CS 334 – Competitive Programming
 CS 334L – Competitive Programming Lab
 CS 333/DES 327 – Introduction to Human Computer Interaction
 CS 335/CE 466 – Introduction to Large Language Models
 CS 343/CE 465 – Graph Data Science
 CS 451 – Computational Intelligence
 CS 363 – Networks & Collective Behavior
 CS 352/CE 368 – Introduction to Reinforcement Learning
 CE 376/CS 358 – Geographic Information Systems
 EE 366/CE 366/CS 380 – Introduction to Robotics

GRADUATING REQUIREMENTS:

A major in BS Computer Science requires completion of (36) courses and a minimum of 130 credit-hours of coursework, with a minimum CGPA of 2.33, as shown in the table below:

Requirements	Course Category	Number of Courses to complete
Habib Liberal Core ¹	University Core	10 ¹
CS Requirement (4 courses + 1 Field Practice ²)	Design Your Habib Experience	1
	Digital Logic & Design	1
	Kaavish I (Final Year Capstone Project)	1
	Kaavish II (Final Year Capstone Project)	1
	Khidmat (Field Practice)	Non-credit component ²
Foundation courses ³ (3 courses)	Algorithmic Problem Solving	1 ¹
	Data Structures and Algorithms	1
	Discrete Mathematics	1
Kernel courses ³ (9 courses)	Database Systems	1
	Object Oriented Programming and Design Methodologies	1
	Data Structures II	1
	System course	1
	Nature of computation	1
	Operating systems	1
	Algorithms: Design and Analysis	1
	Software Engineering	1 ¹
	Artificial Intelligence	1
CS Electives	Electives	5
Natural Science	Natural Science Elective	2
Mathematics	Required Math courses	5 ¹
Professional Practices	Professional Practice Elective	1
Free Electives	Free Electives	4 ⁴
	Over All	36 (minimum)

¹ Three courses double counted in Habib Liberal core as well as one each in Mathematics, CS Foundation and CS Kernel courses.

² Khidmat is a zero credit-hour additional course requirement to be ideally taken in summer semester 2. Khidmat aims for students to engage in society and make meaningful contributions to it in a manner that utilizes their computer science skills. Students engage in fieldwork immersing themselves in an environment that exposes them to real life challenges/issues occurring in society and make a positive impact.

³ Students must obtain a minimum grade of C+ in each CS Foundation and Kernel course.

⁴ The number of free electives attempted must be such that the total number of credit-hours is at least 130. Any course offered at the university, including CS electives, can be counted as a free elective. The quantity of the free electives shown above is for guidance only.

Note 1 - For CS 101, the University's first semester policy applies to only those students attending it in their first semester. Otherwise, the unified grade base applies.

Note 2 - CS Foundation and Kernel courses must be cleared with an earned grade of C+ or higher unless some other grading scheme, e.g., first-semester policy, applies.

OPTION FOR PURSUING MINOR(S):

A student pursuing CS as a major may complete one or more minors currently offered by the University during the course of their four-year academic journey. This may be done by using the free-elective space provided in the 4-year grid. The maximum allowable course load per semester is twenty (20) credit-hours. However, a student in consultation with the Office of Academic Systems & Registrar may be allowed to take additional course load, subject to the University's Course Overload Policy.

4-YEAR GRID:

